

3.5.2 Forces and work

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 4			
A force is a push or pull that acts on an object due to the interaction with another object. All forces between objects are either: • contact forces – the objects are physically touching • non-contact forces – the objects are physically separated.	Examples of contact forces include friction, air resistance, tension and normal contact force. Examples of non-contact forces are gravitational force, electrostatic force and magnetic force.	6.5.1.2	4.6.1.1
Outcome 5			
When a force causes an object to move through a distance, work is done on the object.	Calculations involving work are not required.	6.5.2	4.6.1.3
Work done against the frictional forces acting on an object causes a rise in the temperature of the object.	Suggested activity for TDA Investigate how different surfaces affect the amount of friction on a moving block.	6.5.2	4.7.1.10